Adaptive Sustainable Tech-enabled Robotic Assistant (A. S. T. R. A.)

Presented By

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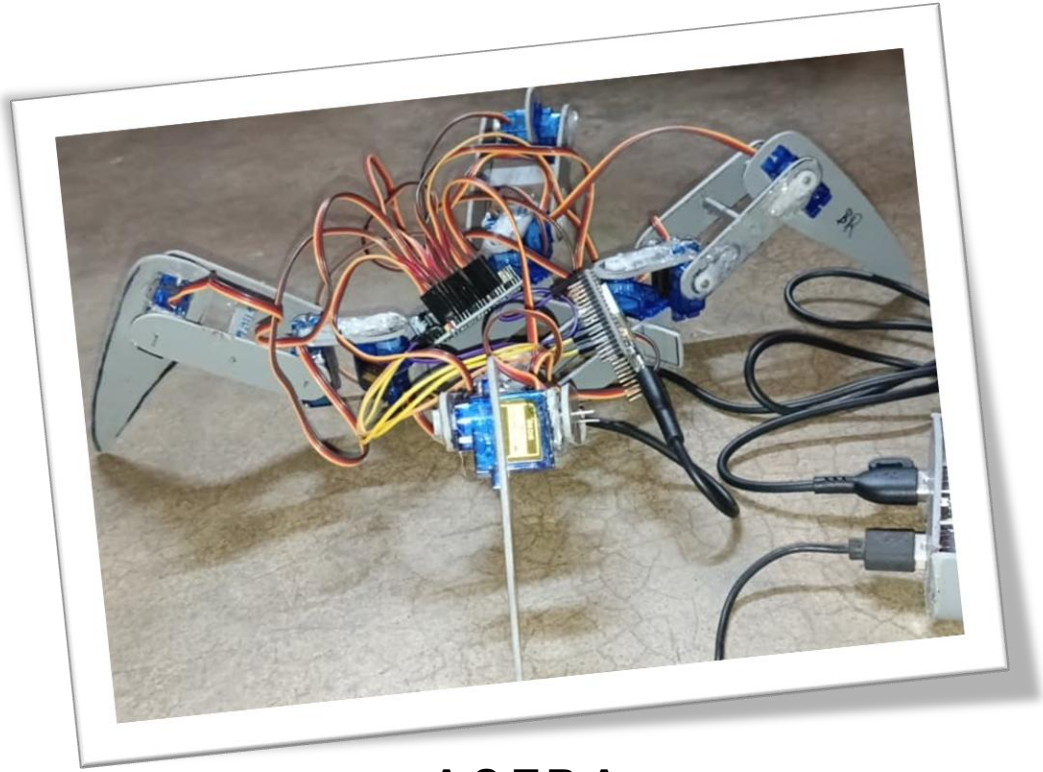

A.S.T.R.A.

***A**daptive **S**ustainable **T**ech-enabled **R**obotic **A**ssistant*

Moving Beyond Controlled Laboratory Settings

Project Overview: A.S.T.R.A. is a compact quadruped locomotion platform engineered for stable movement across uneven terrains. Scope includes integrating a low-cost hardware framework with real-time sensing, hybrid energy harvesting, and seamless human-machine interaction.

Remote Surveillance

- Real-time monitoring in remote or hazardous locations.
- Integrated multi-spectral imaging and motion detection.
- Data transmission via secure, low-latency networks.

Light Industrial Inspection

- Non-destructive testing (NDT) of pipelines and infrastructure.
- Autonomous navigation for routine maintenance checks.
- Environmental sensor integration for leak detection and air quality monitoring.

Advanced Educational Platforms (STEM)

- Hands-on learning for robotics, AI, and control systems.
- Modular hardware for customizable experimentation and research.
- Curriculum support for engineering, computer science, and physics.

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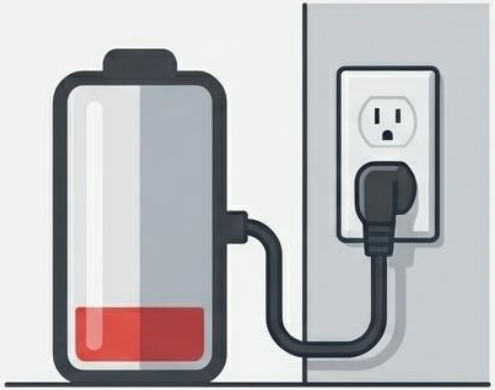
Literature Review

The Terrain and Battery Trap

Mobility Constraints

Conventional wheeled mobile robots are severely limited by flat-terrain dependencies.



Energy Dependency

Current systems are entirely reliant on continuous centralized charging infrastructure.

The Result: Reduced deployment flexibility and compromised operational endurance, particularly in remote, constrained, or energy-scarce environments.

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Identifying the Gaps in Small-Scale Robotics

Problem Statement

Current Limitations

Low-cost robotic platforms lack intelligent control interfaces, real-time monitoring, and adaptability to unstructured environments.

Growing Market Need

A demand for sustainable, IoT-enabled products that integrate renewable energy and offer intuitive remote interaction.

The Problem: Existing solutions combining mobility, sustainability, and intelligent control are currently too expensive, complex, or unsuitable for small-scale deployment.

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Engineering a Sustainable, IoT-Enabled Solution

Methodology

Locomotion

Design a compact ESP32-driven quadruped platform capable of stable, terrain-adaptable movement.

Sustainability

Implement a hybrid energy harvesting system utilizing RF rectennas and miniature solar cells.

Awareness

Integrate real-time sensing (HC-SR04 Ultrasonic, MPU6050 IMU) for navigation and stability.

Control

Develop a low-latency Blynk Dashboard interface for live telemetry, remote navigation, and diagnostics.

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System Modelling

System Modeling and Working Principle

Power Flow

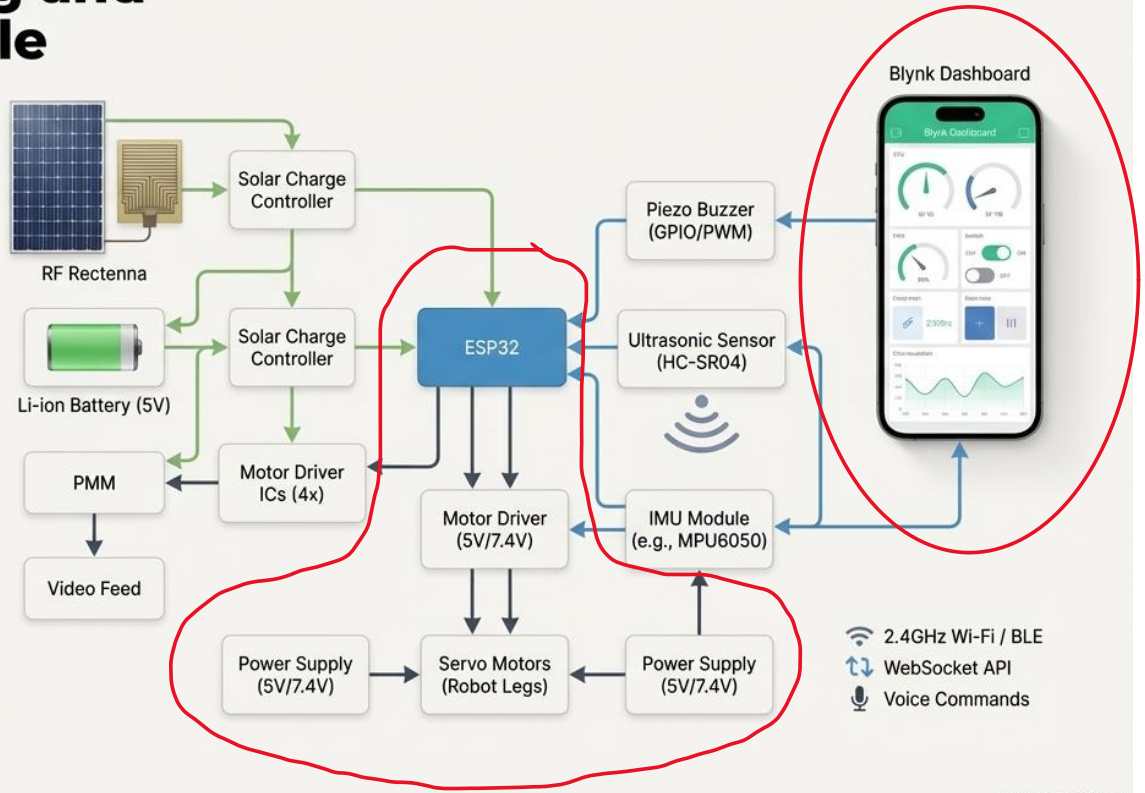
Solar/RF inputs and Li-ion batteries feed into precision motor drivers and step-down converters.

Data Flow

The ESP32 acts as the central brain, processing IMU stability data and Ultrasonic spatial data.

Actuation Flow

Processed signals are sent to the 12 individual servo motors to coordinate complex gait cycles.



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Working Principle

The Brain and The Interface

Microcontroller (ESP32)

The core coordinator. Handles simultaneous multi-threaded tasks—calculating quadruped inverse kinematics while managing Wi-Fi data packets.

IoT Control (Blynk Dashboard)

Replaces heavy, traditional web stacks with a lightweight, ultra-low-latency interface.

Capabilities: Enables remote navigation, real-time battery monitoring, sensor visualization, and voice-command integration over 2.4GHz Wi-Fi/BLE.

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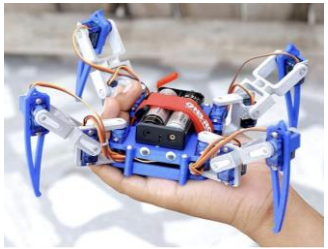
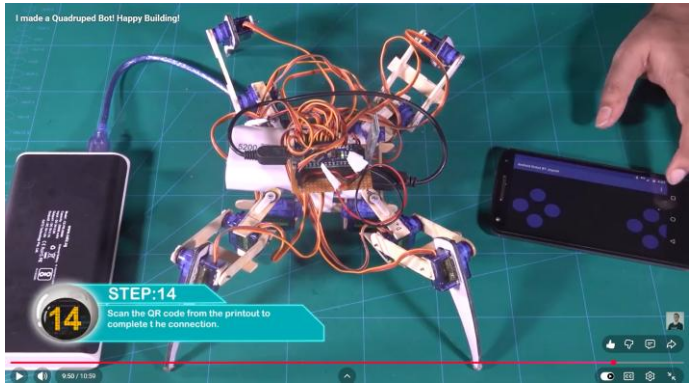
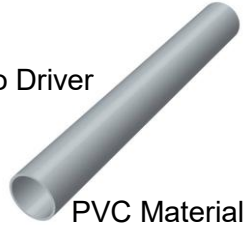




Proto-Product Primary Components

ESP32 Dev Module

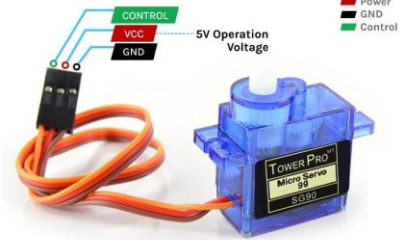
PCA9685 Servo Driver



PVC Material

Type-C Charging Module

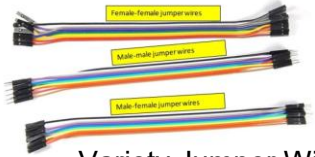
Header Pins



5V Servo Motors

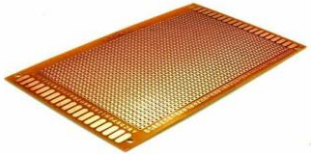
DC-DC Buck Converter

Bread-Board Mini



Variety Jumper Wires

Li+ ion battery
3.7V, 2200mAh



PCB Development
Vero Board

Secondary Components If Advanced

ESP32 Cam

Solar Panel

RF Antenna

IMU Sensor

BMS module

Solar Charge Controller

Raspberry Pi Zero

Joystick

Road Map

Week 1: Literature & Scope

Contextual research and defining hardware requirements.

Week 2: Design & Modeling

3D CAD modeling, PCB planning, and system architecture.

Weeks 3-4: Implementation

3D printing, soldering, and mechanical assembly.

Week 5: Testing & Calibration

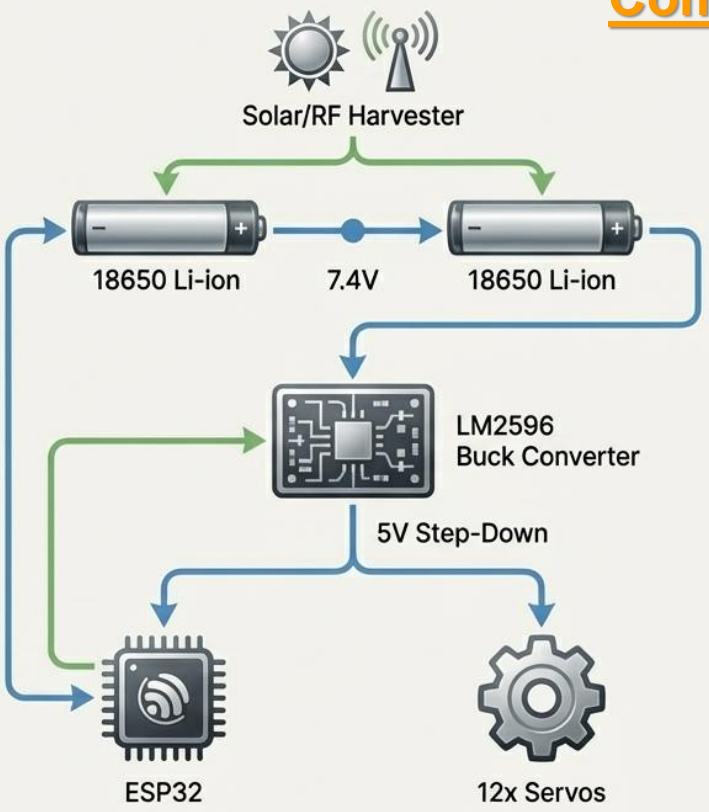
Servo initialization, IMU balancing, and Blynk IoT integration.

Week 6: Documentation

Final performance analysis and project presentation.

Power Management and Hybrid Sustainability

- **Primary Power:** Two 18650 Li-ion batteries connected in series to generate 7.4V.
- **The Critical Bridge:** The LM2596 Buck Converter steps down the 7.4V safely to 5V. This is essential to protect the logic board while delivering the high current required by the servos.
- **Hybrid Harvesting:** An integrated miniature solar cell and RF rectenna capture ambient energy to supplement battery power, extending idle-time endurance.



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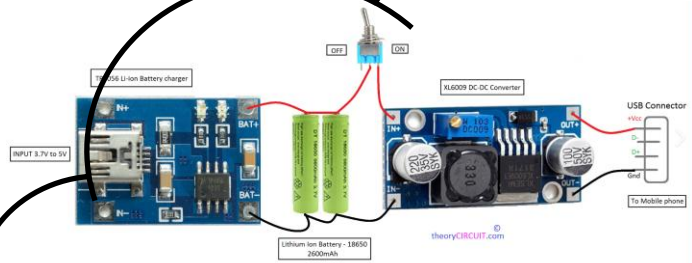
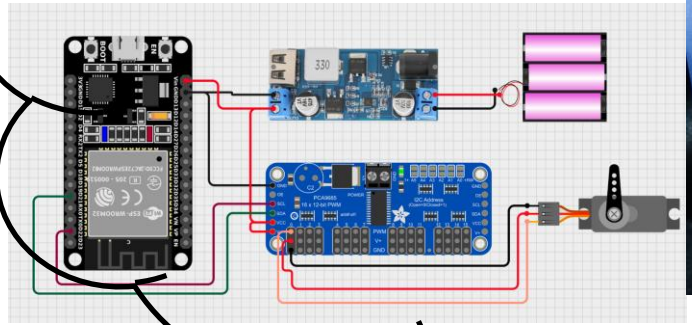
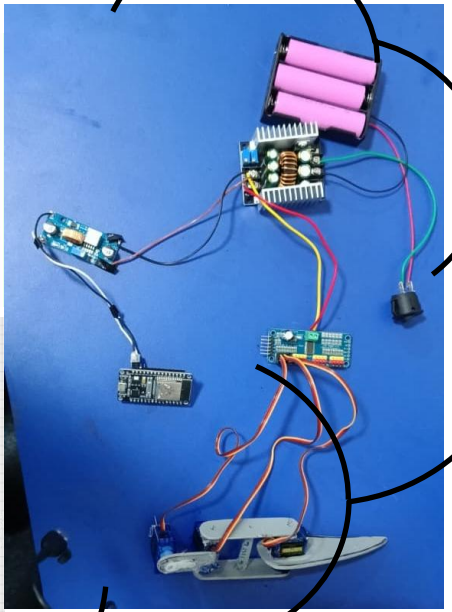
Supply Unit:

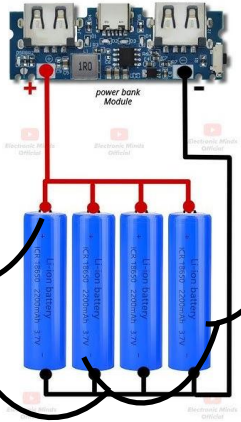




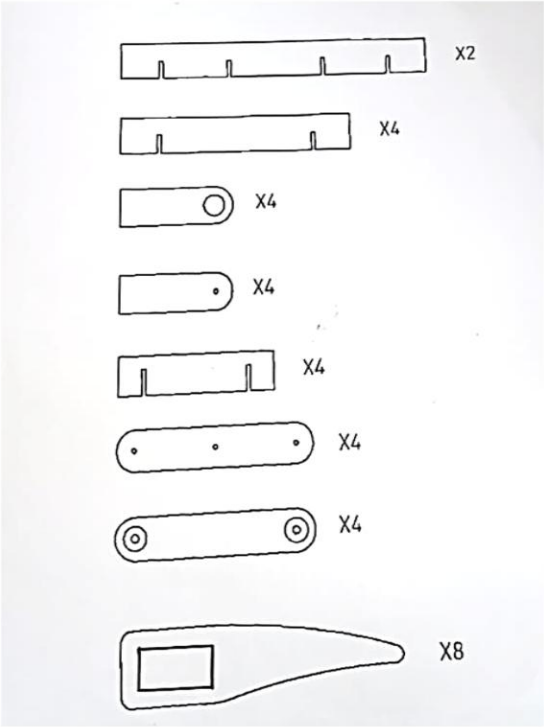


Present Supply Unit:

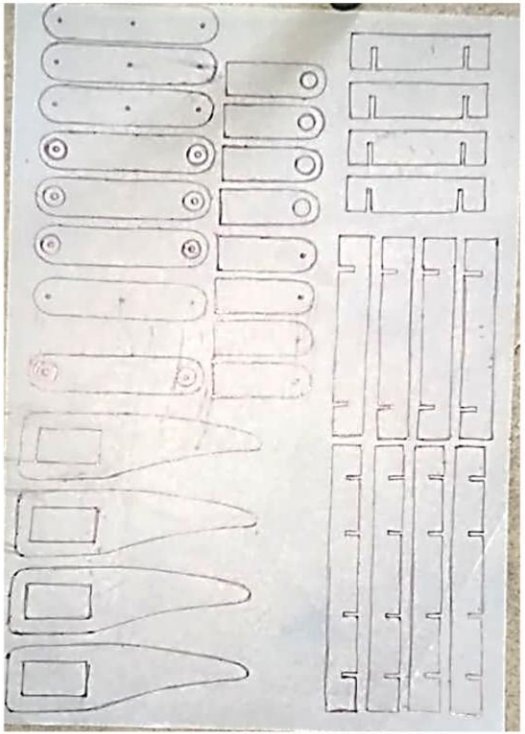
Expected Supply Unit:



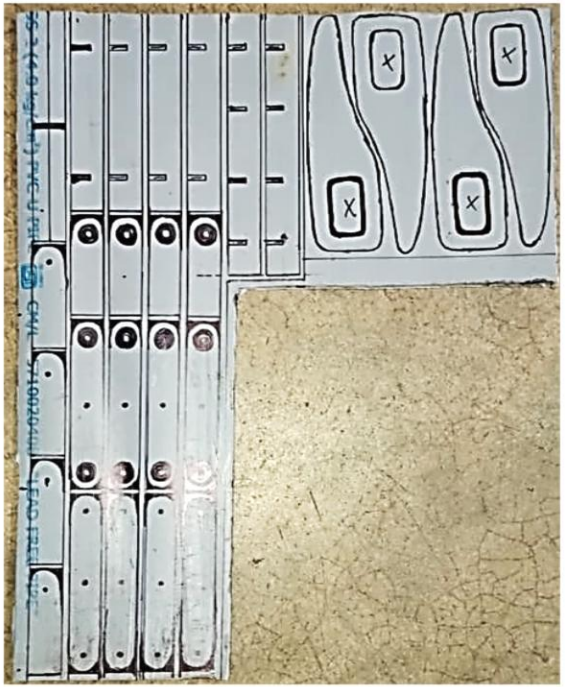
Designing



Printable



Trace



Drawable

Arm Architecture

Actuation and Spatial Awareness

Locomotion

12x SG90 Mini Servos. Chosen for their lightweight plastic gears, perfectly balancing cost-effectiveness with the torque needed for a lightweight 3D-printed PLA/SLA chassis.

Stability

MPU6050 IMU constantly monitors tilt and pitch, allowing the ESP32 to dynamically adjust leg positioning on uneven terrain.

Navigation

HC-SR04 Ultrasonic Sensor acts as the primary optical avoidance system, actively preventing collisions during autonomous or remote operation.

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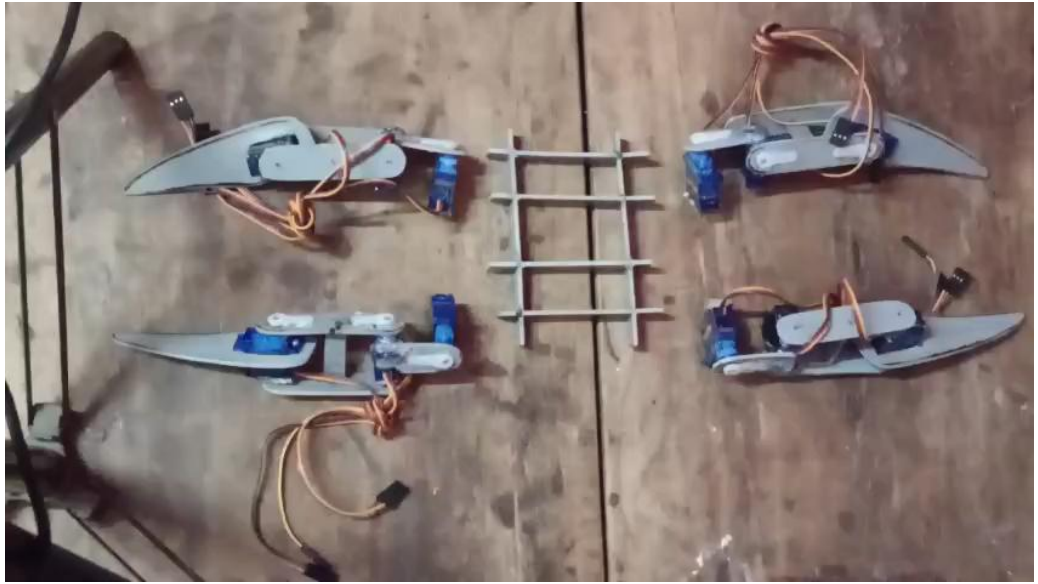
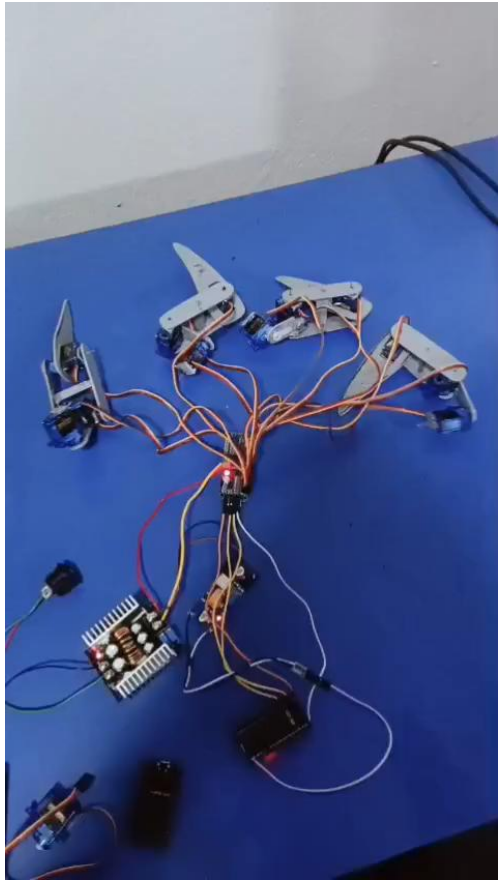
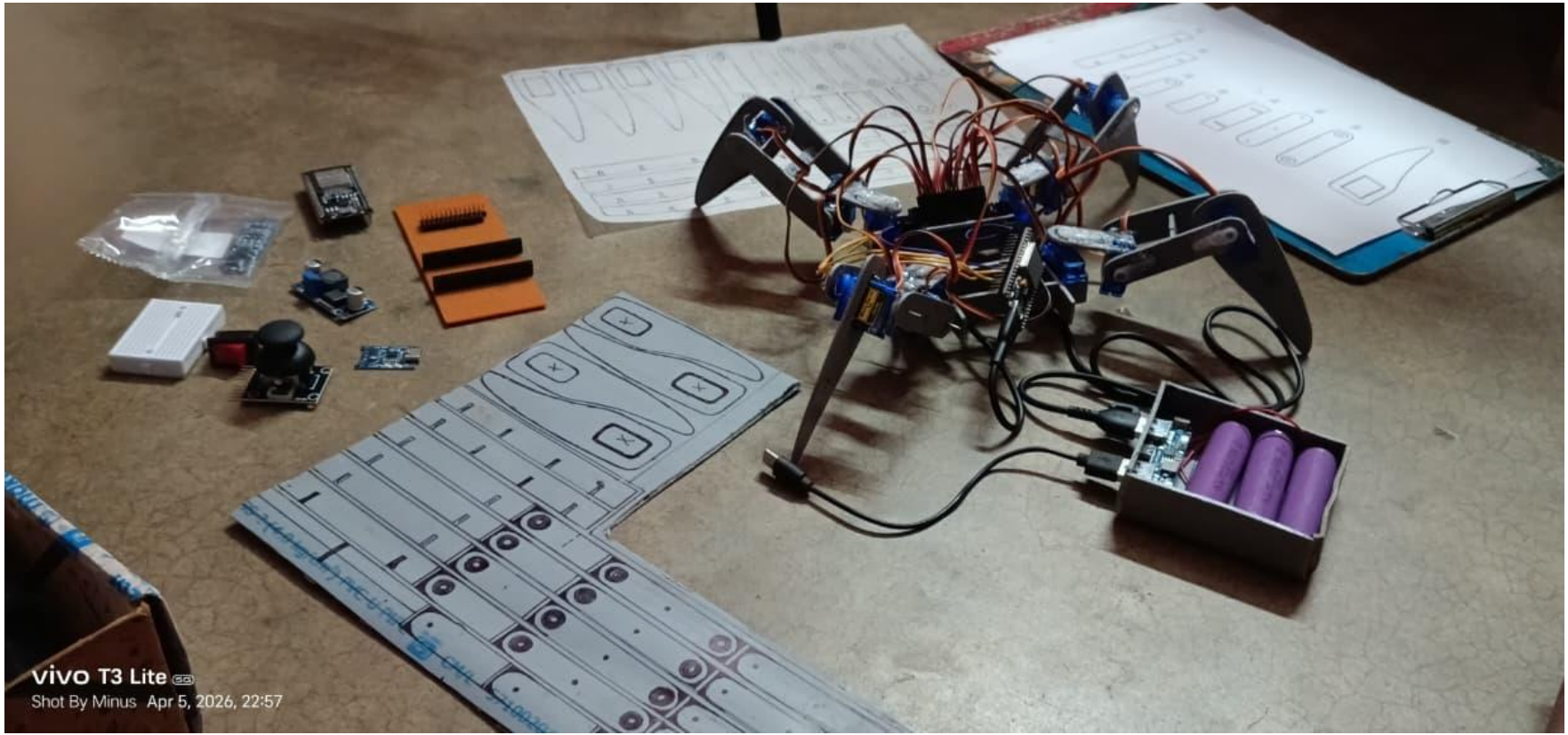
Demo





Circuits



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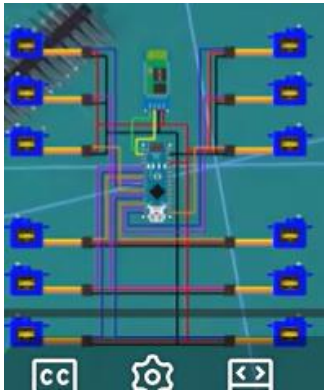
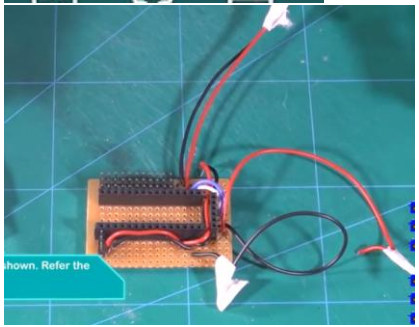
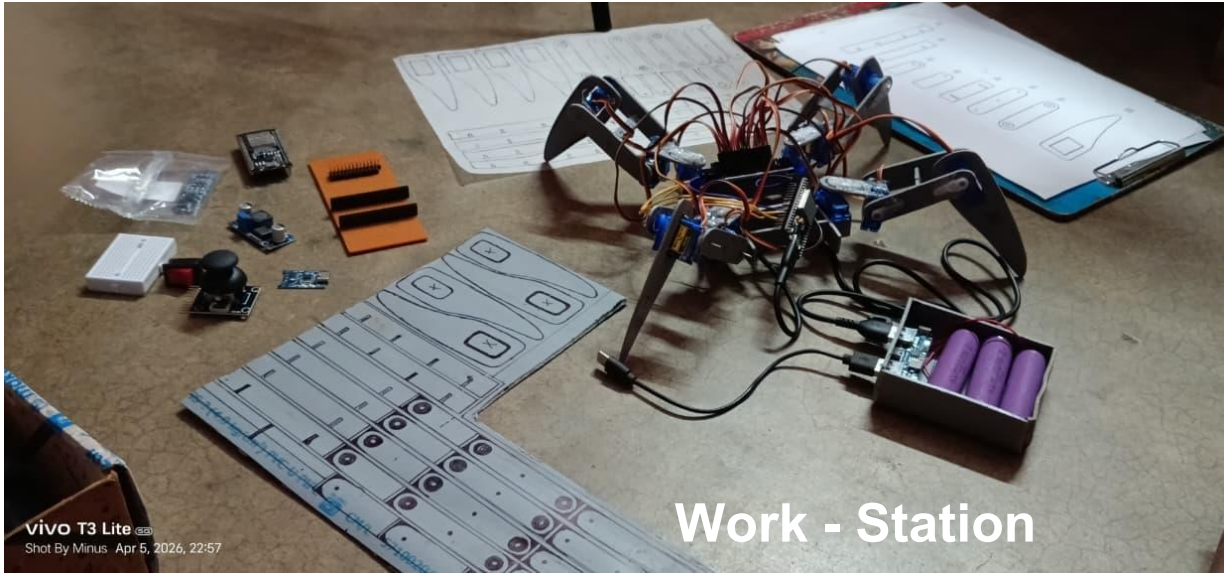
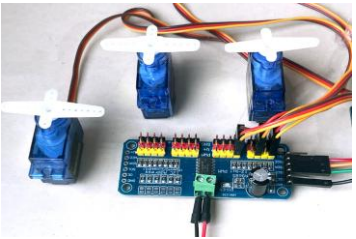


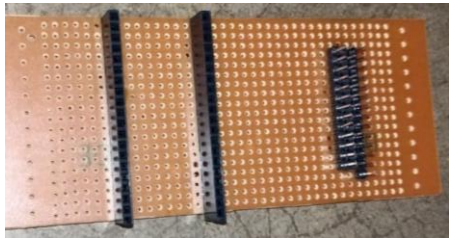


Custom Driver by Creators





Currently using PCA Servo Motor Driver



Our Custom Driver in Progress

Acknowledgements and Project Foundations

- **A.S.T.R.A. Research Foundation:** Heavily drawn from the framework established by Manas R. Das, Bhaktishree Vibek, Raj Tripathy, and Chinmayee Moharana.
- **Open-Source Hardware Communities:** Execution methodologies inspired by the robust open-source robotics communities (Skynet Robotics, SPARKLERS).
- **Final Outcome:** A market-ready, highly adaptable prototype that successfully bridges the gap between complex robotics and accessible, sustainable STEM education.

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